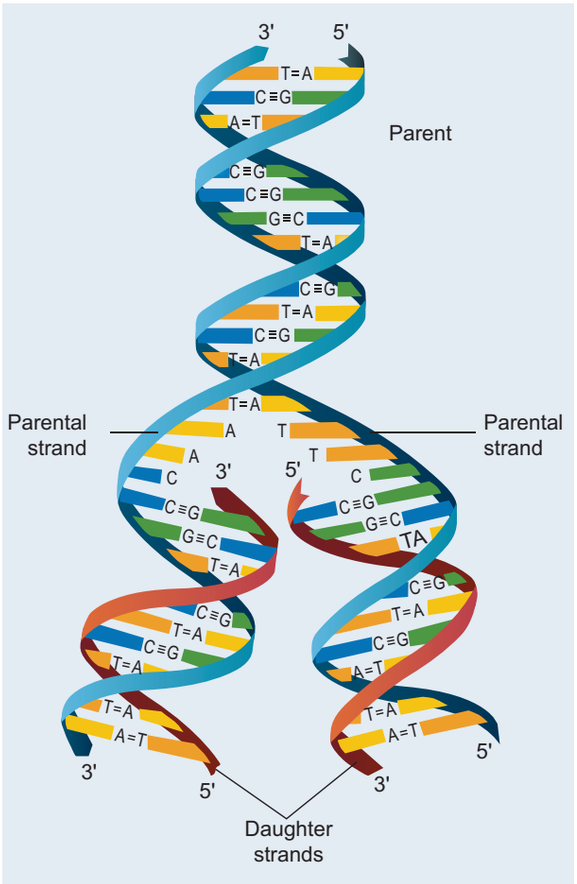


Fig. 5.4 Semiconservative DNA replication

complementary fragments. Finally two new molecules are formed which consist of both old and new fragments.

Semi-conservative replication was proposed by Watson and Crick in 1953. This mechanism of replication is based on the DNA model. They suggested that the two polynucleotide strands of DNA molecule unwind and start separating at one end. During this process, covalent hydrogen bonds are broken. The separated single strand then acts as template for the synthesis of a new strand. Subsequently, each daughter double helix carries one polynucleotide strand from the parent molecule that acts as a template and the other strand is newly synthesised and complementary to the parent strand **(Fig. 5.4)**.

5.8.1 Experimental proof of DNA replication

The mode of DNA replication was determined in 1958 by Meselson and Stahl. They designed an experiment to distinguish between semi conservative, conservative and dispersive replications. In their experiment, they grew two cultures of *E.coli* for many generations in separate media. The 'heavy' culture was grown in a medium in which the nitrogen source (NH_4Cl) contained the heavy isotope ^{15}N and the 'light' culture was grown in a medium in which the nitrogen source contained light isotope ^{14}N for many generations. At the end of growth, they

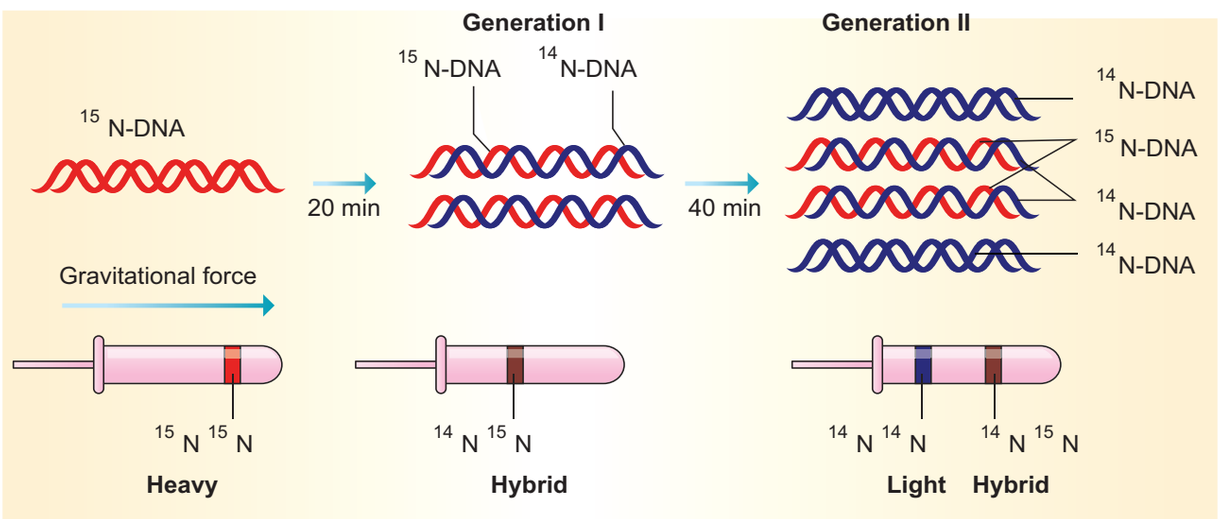


Fig. 5.5 Meselson and Stahl experiment to support semiconservative mode of DNAreplication

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